

Derivation of Radial and Spherical Bessel Basis Functions for DimeNet

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Schrödinger Equation

The derivation in the DimeNet paper begins with the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m}\nabla^2\Psi(\mathbf{r}) + V(\mathbf{r})\Psi(\mathbf{r}) = E\Psi(\mathbf{r})$$

In the free-particle region ($|\mathbf{r}| < c$), the authors of DimeNet simplify the potential $V(\mathbf{r})$ by setting it to zero. Outside the cutoff radius c , they define $V(\mathbf{r}) = \infty$, which enforces the crucial boundary condition $\Psi(|\mathbf{r}| = c) = 0$. This simplification leads to the 3D Helmholtz equation:

$$\nabla^2\Psi(\mathbf{r}) + k^2\Psi(\mathbf{r}) = 0, \quad k^2 = \frac{2mE}{\hbar^2}$$

Laplacian in Spherical Coordinates

To align with the inherent spherical geometry of molecular interactions, the authors then switch to spherical coordinates (d, α, ϕ) , where:

$$d = |\mathbf{r}|, \quad \alpha \in [0, \pi], \quad \phi \in [0, 2\pi]$$

In this coordinate system, the Laplacian operator transforms to:

$$\nabla^2 = \frac{1}{d^2} \frac{\partial}{\partial d} \left(d^2 \frac{\partial}{\partial d} \right) + \frac{1}{d^2 \sin \alpha} \frac{\partial}{\partial \alpha} \left(\sin \alpha \frac{\partial}{\partial \alpha} \right) + \frac{1}{d^2 \sin^2 \alpha} \frac{\partial^2}{\partial \phi^2}$$

Separation of Variables

To solve the Helmholtz equation in spherical coordinates, the authors assume separability of the wave function:

$$\Psi(d, \alpha, \phi) = R(d) \Theta(\alpha) \Phi(\phi)$$

Substituting this into the Helmholtz equation and dividing by $R\Theta\Phi$, the terms can be rearranged:

$$\frac{1}{R} \frac{1}{d^2} \frac{d}{dd} \left(d^2 \frac{dR}{dd} \right) + \frac{1}{\Theta} \frac{1}{d^2 \sin \alpha} \frac{d}{d\alpha} \left(\sin \alpha \frac{d\Theta}{d\alpha} \right) + \frac{1}{\Phi} \frac{1}{d^2 \sin^2 \alpha} \frac{d^2 \Phi}{d\phi^2} + k^2 = 0$$

Multiplying through by d^2 allows for the separation of variables into a radial part and an angular part:

$$\underbrace{\frac{1}{R} \frac{d}{dd} \left(d^2 \frac{dR}{dd} \right) + k^2 d^2}_{=\lambda} = - \left[\underbrace{\frac{1}{\Theta \sin \alpha} \frac{d}{d\alpha} \left(\sin \alpha \frac{d\Theta}{d\alpha} \right)}_{=-\lambda} + \frac{1}{\Phi \sin^2 \alpha} \frac{d^2 \Phi}{d\phi^2} \right]$$

By setting the separation constant to $\lambda = l(l+1)$, where l is a non-negative integer, the authors obtain independent equations for the radial and angular components.

Angular Part Derivation

(a) Azimuthal Equation (ϕ)

The equation for the azimuthal component, as derived from the separation, is:

$$\frac{d^2 \Phi}{d\phi^2} + m^2 \Phi = 0$$

The solutions show a periodic dependence on ϕ , which implies that the constant m must be an integer:

$$\Phi(\phi) = e^{im\phi}, \quad m \in \mathbb{Z}$$

(b) Polar Equation (α)

The equation for the polar component, coupled with the separation constants $l(l+1)$ and m^2 , leads to the associated Legendre equation:

$$\frac{1}{\sin \alpha} \frac{d}{d\alpha} \left(\sin \alpha \frac{d\Theta}{d\alpha} \right) + \left[l(l+1) - \frac{m^2}{\sin^2 \alpha} \right] \Theta = 0$$

The solutions to this equation are the associated Legendre functions:

$$\Theta(\alpha) = P_l^m(\cos \alpha)$$

For physically valid solutions, the integer m must satisfy $|m| \leq l$.

Combined Angular Part: Spherical Harmonics

The combined angular part, crucial for representing directional information, is then represented by spherical harmonics $Y_l^m(\alpha, \phi)$, which are the normalized products of the azimuthal and polar solutions:

$$Y_l^m(\alpha, \phi) = P_l^m(\cos \alpha) e^{im\phi} \quad (\text{with normalization})$$

Radial Equation Derivation

The equation for the radial component is the spherical Bessel differential equation:

$$\frac{d}{dd} \left(d^2 \frac{dR}{dd} \right) + (k^2 d^2 - l(l+1)) R = 0$$

Its general solution involves two types of spherical Bessel functions:

$$R(d) = a_l j_l(kd) + b_l y_l(kd)$$

To ensure the physical regularity of the wave function at the origin ($d = 0$), where $y_l(kd)$ is singular, the authors set $b_l = 0$, selecting only the spherical Bessel function of the first kind:

$$R(d) = a_l j_l(kd)$$

Derivation of Radial Bessel Basis $\tilde{e}_{\text{RBF},n}(d)$

For the purely radial basis function, the authors consider the case where the angular momentum quantum number $l = 0$. In this specific instance, the spherical Bessel function of the first kind simplifies to:

$$j_0(x) = \frac{\sin x}{x}, \quad \text{leading to } R(d) = a_0 \frac{\sin(kd)}{kd}$$

Applying the boundary condition $\Psi(d = c) = 0$, which means $R(c) = 0$, the authors determine the allowed values for k :

$$j_0(kc) = 0 \Rightarrow \frac{\sin(kc)}{kc} = 0 \Rightarrow \sin(kc) = 0$$

This condition is met when kc is an integer multiple of π :

$$kc = n\pi, \quad n = 1, 2, 3, \dots$$

From this, the specific wave numbers are $k = \frac{n\pi}{c}$.

Substituting this back into the expression for $j_0(kd)$ and applying a normalization factor, the authors define the Radial Bessel Basis as:

$$\tilde{e}_{\text{RBF},n}(d) = \sqrt{\frac{2}{c}} \cdot \frac{\sin\left(\frac{n\pi}{c}d\right)}{d}$$

Derivation of 2D Spherical Fourier-Bessel Basis $\tilde{a}_{\text{SBF},ln}(d, \alpha)$

For the 2D spherical Fourier-Bessel basis, the authors fix $m = 0$ (removing azimuthal dependence) and combine the radial and angular components:

$$\Psi(d, \alpha) \propto j_l(kd) Y_l^0(\alpha)$$

Here, $Y_l^0(\alpha)$ is the spherical harmonic for $m = 0$, which depends only on the polar angle α :

$$Y_l^0(\alpha) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \alpha)$$

Enforcing the boundary condition $j_l(kc) = 0$, the authors find that for each l , there is a discrete set of roots z_{ln} (the n -th root of the l -th order spherical Bessel function $j_l(z)$). This determines the wave number k for each l, n pair:

$$k = \frac{z_{ln}}{c}$$

To ensure orthonormality, the authors normalize this basis, resulting in the 2D Spherical Fourier-Bessel Basis used in DimeNet:

$$\tilde{a}_{\text{SBF},ln}(d, \alpha) = \sqrt{\frac{2}{c^3 j_{l+1}^2(z_{ln})}} \cdot j_l\left(\frac{z_{ln}}{c}d\right) \cdot Y_l^0(\alpha)$$

This basis effectively captures both distance and angular information, critical for directional message passing.